

## Motifs Clustering in Time Series Data

### Task Description:

The objective of this task is to identify and analyze recurring patterns (motifs) within the provided signal data. These motifs may vary in both length and pattern shape, and may also exhibit variations due to noise or signal distortion (Figure 1).

In the first stage, you are required to extract candidate motifs from the signal. As an initial approach, an amplitude threshold of 110 may be used to segment the signal and detect potential motif regions; however, this value can be adjusted as needed to improve detection accuracy depending on the signal characteristics. You can ignore the steady patterns around 120.

Following extraction, the identified motifs should be clustered and grouped based on their underlying similarities. This grouping process should consider account relevant features such as **shape and temporal length**. Appropriate similarity measures and clustering techniques should be selected to ensure meaningful grouping. You are **free to apply any appropriate preprocessing techniques and methodological approaches** to improve the motif extraction and clustering.

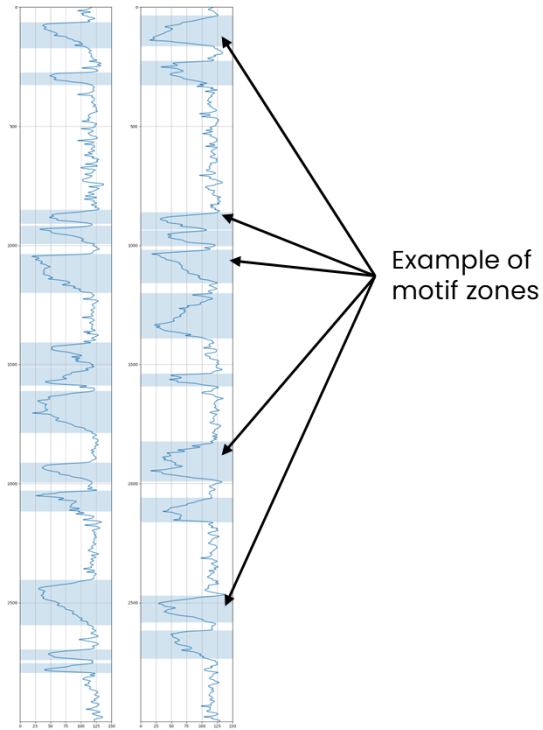
When considering motif length during clustering, segments with similar or close lengths should generally be grouped together. Minor variations in length can be tolerated within the same cluster; however, motifs that differ significantly in length should be separated into distinct clusters to preserve the consistency and interpretability of each pattern group.

The overall goal is to organize the extracted motifs into coherent clusters, where each cluster represents a distinct pattern type (Figure 2). The final output should clearly reflect both the detected motifs and their corresponding group assignments, along with a detailed explanation of the methodology used.

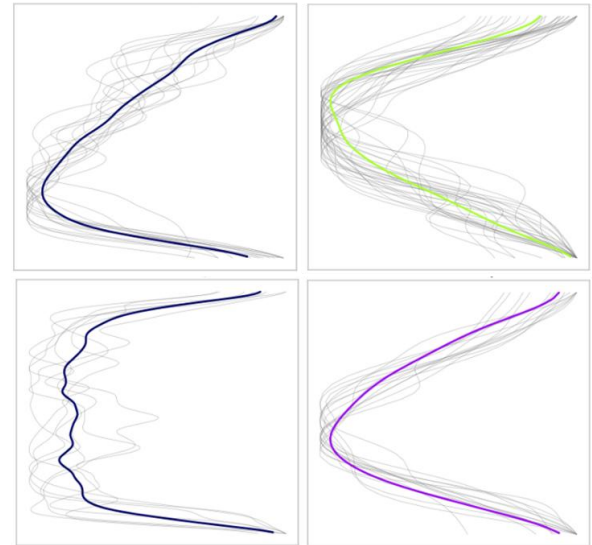
Please send the presentation, source codes, and datasets with your name and surname (for zip file) before the deadline for the review. The source codes can be shared via GitHub or as a zip file.

Best of luck.

Link to data: [signals\\_data.csv](#)



**Figure 1.** Examples of two signal with the blue zones of different motifs that need to be clustered.



**Figure 2.** Example of generalized motifs (bold colored line) with the extracted motifs from signals at the background (lines in grey).

### Dataset Overview:

The dataset is delivered in CSV format and consists of 500 columns, each corresponding to an individual signal. The first row contains the header information for all columns.

For visualization, it is recommended to present the data in a vertical layout, where the y-axis represents the index and the x-axis represents the signal values.

### Expected Deliverables:

You are expected to submit your prepared **scripts** and a **presentation** (for maximum 25 min) that outlines your steps, such as methodology, model performance, its evaluation, and potential improvements.